AI Room Transformer

Your personal space decorator, powered by AI

Cascadia AI Hackathon 2026 · Team of 4 · 8-hour build

Sponsors: Box · AWS · Apify

Box — file storage + moodboard output	AWS Bedrock — vision AI	Apify — product scraping

1. The Concept

Most people have a space that needs love — a bare apartment wall, a cluttered corner, a room that never quite comes together. The problem isn't money or taste; it's not knowing where to start or what to buy.

AI Room Transformer solves this in 90 seconds. Upload photos of any room, describe your style and budget, and the app returns a complete transformation plan: what needs fixing, what to buy, where to buy it, and how it all fits together — saved as a downloadable moodboard in Box.

The one-line pitch

Upload photos of your room + set your vibe + set your budget → get a shoppable, AI-curated transformation plan in under 2 minutes.

2. The Problem Worth Solving

The blank wall problem	People spend months staring at bare walls because choosing art, furniture, or decor feels overwhelming without a starting point.
The budget trap	Interior designers charge $150–$500/hr. Budget shoppers scroll endlessly through Wayfair with no filter for their actual aesthetic.
The mismatch problem	Items bought separately rarely work together. Without a cohesive vision, rooms feel random even after spending money.
The maintenance blind spot	Small visible issues — bad lighting, clutter, poor layout — silently drag down how a space feels, but nobody flags them.

3. How It Works

1

Upload photos

User uploads 1–4 room photos to Box. The app stores them in a per-session folder with metadata tags.

2

Set your vibe

User picks a style tile (cozy, minimalist, bold, maximalist) and sets a total budget via a slider.

3

AI room analysis

AWS Bedrock vision model analyzes each photo — identifying existing furniture, color palette, lighting quality, dimensions, and visible issues.

4

Product curation

Apify scrapes Wayfair, IKEA, and Amazon for real products matching the user's style and budget. AI ranks and curates a shortlist with running cost tracker.

5

Moodboard output

A polished PDF moodboard is assembled — room photos, selected products, fix list, total cost — and saved to Box with a shareable download link.

4. Technical Architecture

Component	Technology	Role
File storage	Box API	Photo uploads, per-user folders, metadata tagging, moodboard PDF hosting + share li

Component	Technology	Role
Vision AI	AWS Bedrock (Claude)	Room photo analysis — furniture identification, issue detection, color palette extractior
Product data	Apify web scraper	Real-time product scraping from Wayfair, IKEA, Amazon filtered by category and max
AI curator	Claude API (Anthropic)	Takes room JSON + vibe + budget, returns ranked shoppable product list with budget
PDF generator	ReportLab / WeasyPrint	Assembles moodboard collage from room photos + product images + fix list
Frontend	React	Upload UI, style tile selector, budget slider, product cards, moodboard preview
Backend	AWS Lambda + API Gateway	Orchestrates the pipeline: Box → Bedrock → Apify → Claude → PDF → Box

5. Team of 4 — Who Builds What

P1 Box Integration

- Set up Box OAuth + upload endpoint (H1)
- Create per-session folder structure with metadata (H2–3)
- Wire frontend uploads to real Box storage (H4–5)
- Build moodboard PDF + save to Box + share link (H5–6)
- README + GitHub push (H8)

P2 Vision AI Pipeline

- Get AWS Bedrock vision call working on 1 test photo (H1)
- Build full room report JSON: furniture, issues, palette, style (H2–3)
- Batch analyze all uploaded photos (H4)
- Wire vision output into P3 pipeline, end-to-end test (H5–6)
- Demo photo prep + smoke testing (H7–8)

P 3 AI Product Curator

- Apify scraper pulling Wayfair/IKEA products by category + price (H1–2)
- AI curator prompt: room JSON + vibe + budget → product list (H3–4)
- Budget tracker: running total + substitution logic if over budget (H5)
- Trend layer: Apify scrapes Pinterest for style keywords (H6–7)
- Demo script + fallback static data (H8)

P 4 Frontend UI

- Upload UI + vibe style tiles + budget slider with mock data (H1–2)
- Room analysis results card + shoppable product grid (H3–4)
- Wire real AI data end-to-end. Budget meter. Fix list with urgency tags (H5–6)
- Moodboard preview in-app + Box download button (H7)
- Final UI polish + demo run-through (H8)

6. Hour-by-Hour Timeline

Hour	P1 — Box	P2 — Vision AI	P3 — Product AI	P4 — Frontend
H1–2	Box app + OAuth. Upload endpoint live.	Bedrock vision call on test photo. JSON output.	Apify scraper pulling products by category + price	Upload UI + vibe selector + budget slider (mock
H3–4	Folder structure + metadata tags per session.	Full room report JSON: furniture, palette, issues, style.	AI curator prompt returning ranked product list.	Room analysis card + shoppable product grid.
H5–6	Moodboard PDF assembled + saved to Box. Share link.	Wire vision output into P3. End-to-end test.	Budget tracker + over-budget substitution logic.	Wire real data end-to-end. Budget meter + fix lis
H7	Polish + README draft.	Full smoke test with demo photos.	Pinterest trend scraping wired into suggestions.	Moodboard preview in-app. Box download butto
H8	GitHub push + README final.	Bug fixes.	Demo script + fallback data ready.	Final UI polish + demo rehearsal.

7. Team Sync Milestones

End H2	P2 shares room analysis JSON schema with P3 and P4. Both build against this structure immediately — no waiting.

End H3	P3 shares product list JSON shape with P4. P4 builds product cards against the real data structure.
End H5	Full pipeline smoke test: upload 2 photos → room analysis → product list → budget meter → download moodboard from Box.
End H7	Feature freeze. Everything after this point is polish, bug fixes, and demo rehearsal only. No new ideas.

8. Stage Demo Script (90 seconds)

0:00	**Hook**	"Everyone has a room that needs love but no idea where to start or what to buy."
0:15	**Input**	Upload 3 pre-loaded room photos. Select 'cozy minimalist'. Set $300 budget.
0:30	**Analysis**	AI room analysis appears: 'bare walls, harsh overhead light, needs warmth and texture.'
0:50	**Products**	8 shoppable product cards appear — throw blanket $24, floor lamp $67, art print $18... running total: $287.
1:10	**Close**	Click download — moodboard PDF opens from Box. "This is your room, transformed, for $287."

9. Win Conditions

Level	What's Working	Good Enough To?
Minimum (if things break)	Upload 1 photo → AI describes room → 5 hardcoded product cards → Box link works	Show the concept. Explain the vision. Still worth presenting.
Good demo	2 photos → real AI analysis → 8 shoppable picks under budget → budget meter	Competitive. Judges understand the full value.
Full winner	3 photos + vibe + budget → room analysis + fix list + curated products + downloadable Box moodboard	Win best overall + Box prize.

10. Risks & Mitigations

Risk	Level	Mitigation
Box OAuth	HIGH	Takes 30 min if unfamiliar. P1 starts this before anything else — nothing else in hour 1.
Bedrock access	HIGH	Vision model must be enabled in AWS console. Check within the first 10 minutes of the hackathon.
Apify scraping blocked	MED	Wayfair/IKEA may block scrapers. Fallback: pre-scrape 60 products into a static JSON file before the ha
Scope creep	MED	Lock features at H4. Room analysis + product list + moodboard = enough to win. No new ideas after H4.
Moodboard PDF quality	LOW	Keep it simple: text + product names + prices. A clean layout beats a complex collage that breaks.
Demo performance	LOW	Pre-load all demo photos. Do a full run-through at H7. Have static screenshots as a backup.

11. Starter Code — What to Build First

P1 — Box

- `POST /upload endpoint: receive multipart/form-data, stream to Box using chunked upload API`
- `createSessionFolder(userId): create Box folder structure /sessions/{id}/photos/ and /sessions/{id}/output/`
- `generateShareLink(fileId): return a direct download URL from Box for the moodboard PDF`

P2 — Vision AI

- `analyzeRoom(imageBase64): call Bedrock claude-3-5-sonnet with vision, return structured JSON`
- `JSON schema: { furniture: [], issues: [], colorPalette: [], styleInference: string, dimensionsEstimate: string }`
- `Batch function: analyzeAllPhotos(photoUrls[]) — process in parallel, merge into single room profile`

P3 — Product AI

- Apify actor config: target Wayfair category pages, extract name/price/URL/image, filter by maxPrice param
- curateProducts(roomProfile, vibe, budget): Claude prompt returning top 8 picks as [{name, price, url, reason}]
- Budget tracker: runningTotal(), swapIfOver(item) — suggest cheaper alternative if cart exceeds budget

P4 — Frontend

- VibeSelector component: 4 style tiles (cozy / minimalist / bold / maximalist) with visual preview
- ProductCard component: image, name, price badge, 'Add to plan' button, cumulative budget meter
- MoodboardPreview: grid of selected items + room photos + total cost + Box download button

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